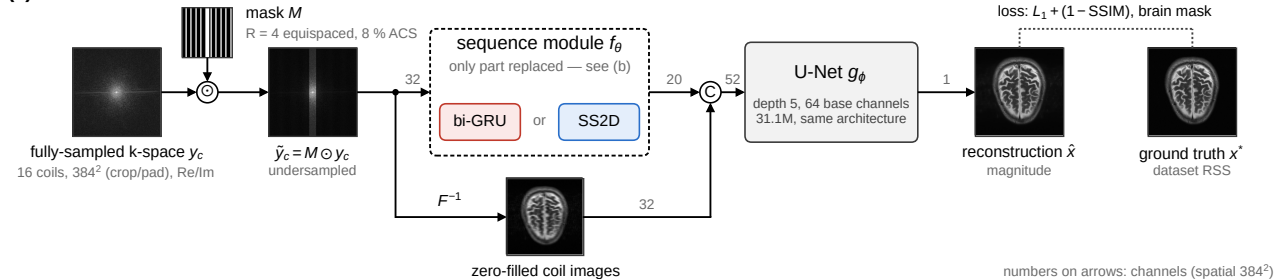
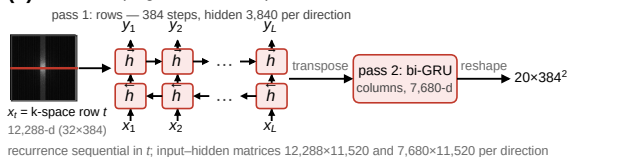


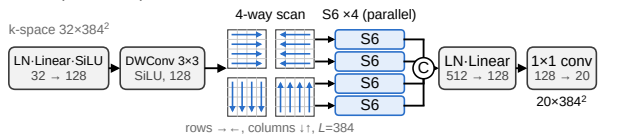
(a) common ETER-Net pipeline (no weight sharing)



(b) bi-GRU (original ETER-Net)



SS2D (controlled)



$$S6: h_t = \bar{A}_t h_{t-1} + \bar{B}_t x_t, \quad y_t = C_t h_t + D x_t, \quad (\Delta_t, B_t, C_t) \text{ from } x_t$$

128 inner channels, $N = 16$; one S6 weight set per direction, shared by all its rows (columns)

34.2M (SS2D module 3.1M), fp16 scan

(c) SS2D (enhanced) — replaces f_θ in (a)

